

SEPTEMBER EXAMINATION – 2025

SUBJECT : BIOLOGY

CLASS – XII

Time Allowed : 03:00 Hours

Maximum Marks : 70

GENERAL INSTRUCTIONS:

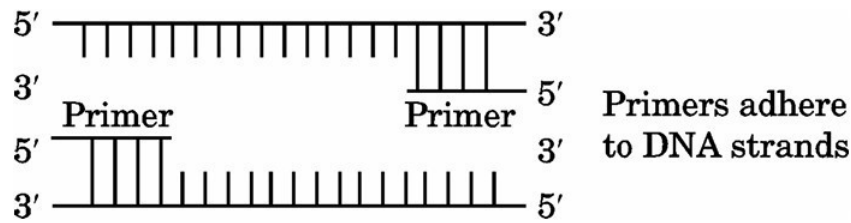
- (i) All questions are compulsory.
- (ii) The question paper has five sections and 33 questions.
- (iii) Section–A has 16 questions of 1 mark each; Section–B has 5 questions of 2 marks each; Section– C has 7 questions of 3 marks each; Section– D has 2 case-based questions of 4 marks each; and Section–E has 3 questions of 5 marks each.
- (iv) There is no overall choice. Answer all 33 questions. However, internal choices have been provided in some questions. A student has to attempt only one of the alternatives in such questions.
- (v) Wherever necessary, neat and properly labeled diagrams should be drawn.

Section – A

Q. No. 1 to 12 are multiple choice questions. Only one of the choices is correct.

Select and write the correct choice as well as the answer to these questions.

1. The primary endosperm nucleus is formed by fusion of which of the following?
 - A. A male gamete and a female gamete
 - B. A male gamete and two polar nuclei
 - C. A female gamete and two synergids
 - D. Two male gametes and an egg cell
2. During the menstrual cycle of a human female, formation of graafian follicle is stimulated by secretion of which of the following gonadotropin hormones?
 - A. Estrogen and progesterone
 - B. FSH and Estrogen
 - C. FSH and LH
 - D. Progesterone and LH
3. The male gametes are formed by:
 - A. Mitotic division of nucleus of vegetative cell
 - B. Meiotic division of nucleus of vegetative cell
 - C. Mitotic division of nucleus of generative cell
 - D. Meiotic division of nucleus of generative cell
4. Short stretches of DNA used to identify complementary sequences in a sample are called
 - A. Probes
 - B. Markers
 - C. Primers
 - D. Minisatellites
5. The experimental proof on the thermal stability of genetic material was first provided by experiments of
 - A. Hershey and Chase
 - B. Meselson and Stahl
 - C. Frederick Griffith
 - D. Jacob and Monod
6. What would be the genotype of parents if the F₁ offspring have the phenotypes in 1 : 1 proportion ?
 - A. Aa × Aa
 - B. AA × AA
 - C. Aa × AA
 - D. Aa × aa

7. Albinism is known to be due to an autosomal recessive mutation. The first child of a couple with normal skin pigmentation was an albino. What is the probability that their second child will also be an albino?
 A. 100% B. 25%
 C. 50% D. 75%
8. Which of the following animals exhibit male heterogamety ?
 (i) Fruit fly (ii) Fowl
 (iii) Human (iv) Honey bee
 (A) (i) and (iii) (B) (ii) and (iv)
 (C) (ii) and (iii) (D) (i) and (iv)
9. What would happen if in a gene encoding a polypeptide of 51 amino acids, codon present at 32 position (UAU) is mutated to UGA ?
 Select the option that indicates the polypeptide with correct number of amino acids formed after mutation.
 (A) A polypeptide of 50 amino acids will be formed.
 (B) A polypeptide of 32 amino acids will be formed.
 (C) Two polypeptides of 20 and 31 amino acids will be formed.
 (D) A polypeptide of 31 amino acids will be formed.
10. Down is syndrome is due to
 (A) NON-DISJUNCTION (B) TRISOMY OF CHROMOSOM E21
 (C) Both (A) & (B) (D) None of these.
11. What is the process of binding a primer to the denatured strand called ?
- 
- (A) Denaturation (B) Annealing
 (C) Renaturation (D) Polymerisation
12. Name the Hormone which is produced in females during pregnancy
 (A) ESTRUGN (B) hPL
 (C) LH (D) Both (A) & (B)

Question No. 13 to 16 consist of two statements – Assertion (A) and Reason (R). Answer these questions selecting the appropriate option given below:

- A. Both A and R are true and R is the correct explanation of A.
 B. Both A and R are true and R is not the correct explanation of A.
 C. A is true but R is false.
 D. A is False but R is true.
13. **Assertion (A):** The ability of the pistil to recognise the pollen is the result of a continuous dialogue between pollen and pistil.
Reason (R): This electrical dialogue allows only compatible pollen to germinate.
14. **Assertion (A):** Some organisms are better adapted to survive in an otherwise hostile environment.
Reason (R): Adaptive ability is inherited and has a genetic basis.
15. **Assertion (A):** DNA chemically is less reactive and structurally more stable when compared to RNA
Reason (R): DNA has 2'-OH group present at every nucleotide, which makes it more stable than RNA.

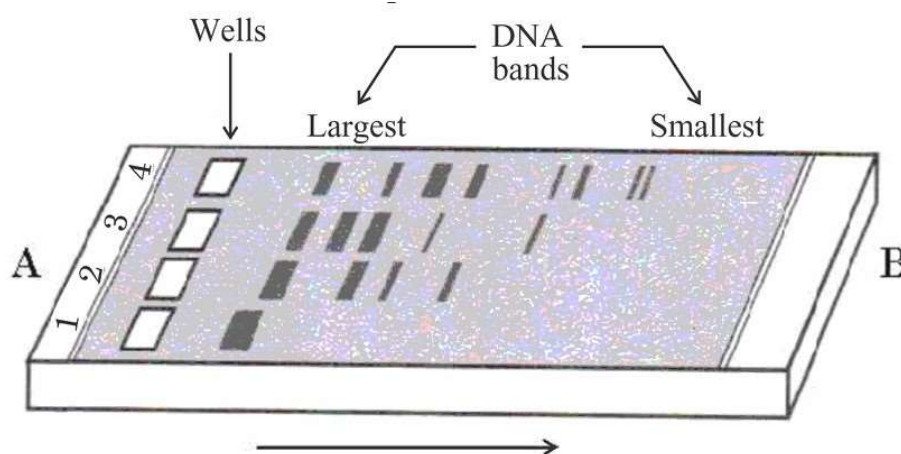
16. **Assertion (A) :** Amino centesis is a process of Foetal Sex Determination.
Reason (R) : Metabolic disorders and other diseases can be diagnosed properly.

SECTION – B

17. During artificial hybridisation it is important to ensure that only desired pollen grains are used for pollination. How is it ensured?
18. How is the rate of initiation of RNA polymerase at a given promoter in a transcriptional unit of prokaryotes regulated ?
19. The table below shows a hypothetical blood report of a patient.

Test description	Observed value	Unit	Reference range
<i>Leucocytes</i>			
Total leucocyte count	1100	Per Microliter	4400-11000
Neutrophils	31	%	55-70
Lymphocytes	25	%	20-40
Basophils	0.5	%	0.5-1
Eosinophils	02	%	1-4
Monocytes	0	%	1-8

- A. Looking at the values suggest which defense mechanism/ immunity is affected and state how this defense mechanism provides immunity.
- B. Name the barrier with least count and enumerate its role in providing immunity.
20. Given below is the diagram representing the observation made for separating DNA fragments by gel electrophoresis. Observe the illustration and answer the questions that follow :



- (a) DNA fragments move in the direction A→B. Explain.
- (b) Name the matrix used and its role in gel electrophoresis.
- (c) Mention how the separated fragments can be visualized for further technical use.
21. Explain Antibiotic Resistance observed in bacteria in the light of DARWIN'S theory of natural selection.

OR

Define convergent evolution. Give suitable examples also.

SECTION – C

22. Suggest a suitable contraceptive device for the following cases with justification.
- (i) Mohini does not want to take the risk of conception and sexually transmitted infections (STIs).
 - (ii) Lalita has two children and does not want any more children.
 - (iii) Geeta wants a contraceptive that she can take till she wants to avoid conception and can resume back to her fertile life without the intervention of the doctor. Also, it should have a lower failure rate.
23. Given below is a figure showing transport of ovum, fertilisation and passage of growing embryo through fallopian tube in a human female. Answer the questions that follow:
- (i) What will be the ploidy of cells shown in (a) and (c) stage in the figure given below?
 - (ii) What will happen if component L as shown in the figure (g) given below does not attach properly to the endometrium?
 - (iii) In a pregnant mother (case X), during early pregnancy, the fertilised egg splits into two embryos at stage C shown in the figure given below, resulting in the formation of twins. Will the genome of cells of these two embryos exhibit variation? Justify your answer.
24. In guinea pigs, black coat colour (G) dominates over white (g) and brown eyes (B) dominate over blue (b). The alleles for coat colour and eye colour are not linked. What will be the probability of the offspring having blue eyes and a white coat if both parents are heterozygous for eye and coat colour? Find the probability using a Punnett square.
25. Define:
- (i) OKAZAKI FRAGMENTS
 - (ii) CO-DOMINANCE
 - (iii) Incomplete Dominance
26. (a) Why do sportspersons often fall victim to cocaine addiction ? Name the plant from which this drug is derived. Also write down the mode of action of this drug.
- (b) Name one drug and its harmful effect obtained from latex of Poppy plant.
27. (a) Write a note on PHENYL KETONURIA.
- (b) Explain Haemophilia & its causes in detail.
28. Explain the law of "Independent Assortment" using a dihybrid cross.

SECTION – D

29. Explain the process of Megasporogenesis with the help of suitable diagrams.
30. Explain with suitable diagram how TH Morgan concluded that genes are located on chromosomes in a linear fashion (with reference to drosophila).

OR

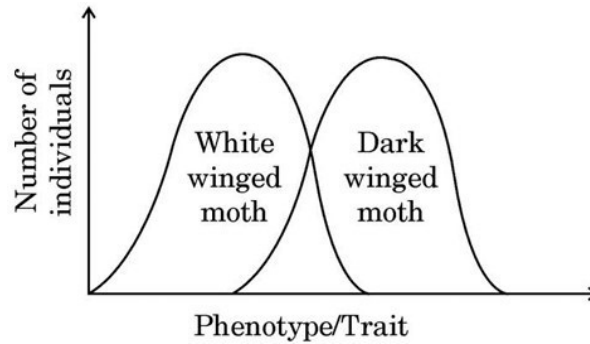
Draw a neat and well labelled diagram of female reproduction system.

SECTION - E

31. (a) Define Histone proteins. Explain the four types of Histone proteins and Histone Octamer with diagram.
- (b) Define the following
- (i) Anti Parallel nature of DNA strands.
 - (ii) Kornberg's Enzyme
 - (iii) Exons

OR

- (a) Explain the process of aminoacylation of tRNA. Mention its role in translation.
- (b) How do ribosomes in the cells act as factories for protein synthesis?
- (c) Given below is a strand of mRNA undergoing the process of translation, what will be the sequence of Amino acids that will be translated? Name the triplet codons that should be added to bring to the end of translation at X.
32. (a) (i)



Natural selection operates in different ways in nature.

- (I) In England after industrialisation, the population of dark winged moths were more favoured than white winged moths. Explain.
- (II) Identify the type of natural selection depicted in the graph above.
- (ii) State Hugo de Vries' theory of mutation. How is it different from Darwin's theory of natural selection ?

OR

Explain the regulation of gene expression in Bacteria using example of LAC-OPERON.

33. (a) (i) Draw a diagram of an enlarged view of transverse section (T.S.) of one microsporangium of an angiosperm and label the following parts :
- | | |
|-------------------|-----------------------------|
| (I) Tapetum | (II) Middle layers |
| (III) Endothecium | (IV) Microspore mother cell |
- (ii) Write the function of Tapetum.
- (iii) Explain the following giving reasons :
- (I) Pollen grains are well-preserved as fossils.
- (II) Pollen tablets are in use by people these days.

OR

- (b) (i) Where does spermatogenesis occur in human testes ? Describe the process of spermatogenesis up to the formation of spermatozoa. 3
- (ii) Trace the path of movement of spermatozoa from the testes upto the ejaculatory duct. 2